

Biology

Chapter 2.1: DNA & RNA

Welcome to AnithUncommon's Sample Resources!

We are assuming you are reading this as a student. So What will **YOU** Expect From This Sample?

In Biology Chapter 2.1, you will explore the language of living organisms that allows the organism's body to properly function

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Kenneth V. to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology, join our nonprofit!

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This sample is made thanks to, Kenneth V.

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be *able* to...

- Describe the structure of DNA & RNA and how they are held together
- Explain the function of both DNA & RNA

Reflective Questions to Consider:

The DNA is known as the coding language of living organisms. Thus:

- What parts of the DNA 'code'?
- What are the differences between DNA and RNA?

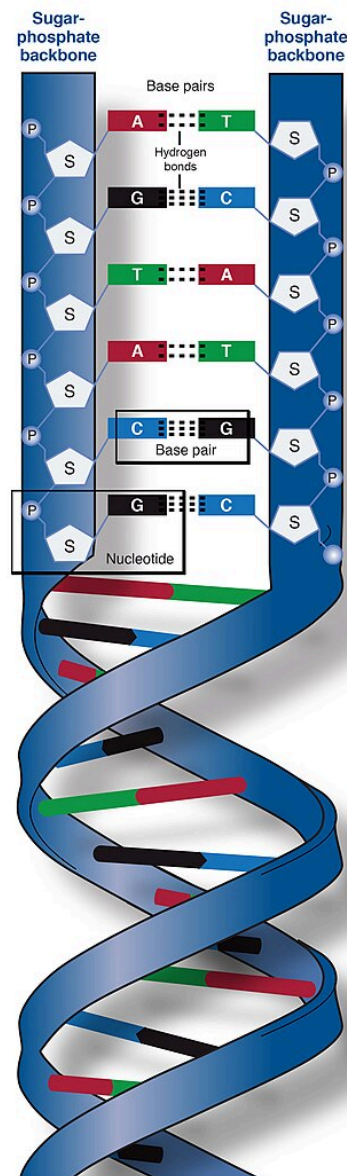
Chapter 2: Genetics	
Tip! • It's good to recall the basics of Organic Chemistry when studying about subjects related to the molecules within living organisms, specifically DNA & RNA in this chapter Activities in Class: • Lesson Breakdown • Kahoot-type quiz	
2.1 . DNA & RNA Deoxyribonucleic Acid (DNA) A. Description → DNA is considered a Polynucleotide (Poly-	Mentor's Notes It cannot be stressed hard enough that the DNA is the genetic

meaning many)

→ A **nucleotide** consists of a **Nitrogenous base**, **Phosphate group**, and a **deoxyribose sugar** (Pentose)

→ The **bonds** between nucleotides that hold them together inside DNA are called **phosphodiester bonds**

B. Structure



code of living organisms. It's important to realize that DNA is most likely found within the nucleus of a cell (found as chromosomes during cell division). Imagine the amount of DNA molecules there are if we have 30-40 trillion animal cells, each molecule unique to code for a specific instruction in one cell.

→ A DNA molecule is made of **two long chains** of polynucleotides that run opposite to each other (**Antiparallel strands**)

→ Each of the branches you see (Labelled AT, GC, TA, etc.) are called **base pairs**. Each base has its own **complementary pair**. They are **hydrophobic**

→ The bases are held together by **strong hydrogen bonds**, keeping them secure

→ The **Sugar-Phosphate backbone** formed from linking nucleotides serves as the **structural framework** of the molecule. It is **hydrophilic**

→ The DNA Molecule is twisted into a **helical shape** to minimize the amount of space it uses and to maximize its **stability**. By twisting it, it also protects the hydrophobic bases with the hydrophilic sugar-phosphate backbone

C. Function

→ The DNA's primary function is to hold **instructions** on how a living organism builds, maintains, and operates itself. It is essentially a blueprint of a living organism

→ It is able to be replicated and passed down through inheritance, it can even **mutate** to form the possibility of evolution

Nitrogenous Bases

A. Description

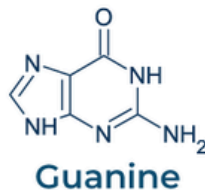
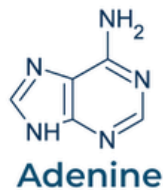
→ There are **four** Nitrogenous bases that are in **DNA** Molecules and **one** specific to **RNA** Molecules

→ As the name suggests, these are **organic molecules** that contain **Nitrogen**

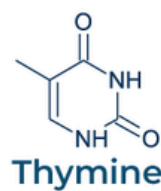
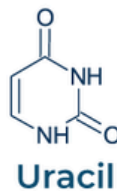
→ The four bases found in DNA are: **Adenine**, **Thymine**, **Guanine**, **Cytosine**, with **Uracil** found in RNA specifically

B. Structure

Purines



Pyrimidines



→ Nitrogenous bases are split into two categories based on the amount of rings they have: **Purines** and **Pyrimidines**

→ Adenine and Guanine are Purines, while Cytosine, Uracil, and Thymine are Pyrimidines

→ **Purines are only complementary to Pyrimidines when pairing,**

→ Adenine pairs with Thymine, Guanine pairs with Cytosine, while Uracil replaces Thymine in RNA

Ribonucleic Acid (RNA)

A. Description

→ RNA is considered a **polynucleotide** too

→ It has a similar structure yet its sugar-phosphate uses a different sugar known as **Ribose** (Containing

an extra oxygen atom than Deoxyribose)

→ Instead of being double-stranded, it consists of only a **single strand**

→ It comes in many forms as: **mRNA, tRNA, rRNA**, etc.

B. Function

→ RNA plays a crucial role in the creation of proteins, particularly using mRNA, tRNA, and rRNA

→ mRNA comes from DNA, its job is to **encode proteins**

→ The mRNA then gets transported to a **Ribosome** where tRNA brings the amino acids that are complementary to a gene code (three nitrogenous bases together) until the rRNA links the amino acids together forming peptide bonds to create a protein

Comparison between DNA & RNA

	DNA	RNA
Structure	Two strands of Nucleotides in a helical shape	One strand of nucleotide in a straight chain
Bases	Adenine, Thymine, Guanine, Cytosine	Adenine, Uracil, Guanine, Cytosine
Function	Contains the genetic code for a living organism to build, maintain, and operate itself (Long term storage)	Carries the genetic code to ribosomes for protein synthesis (Transfers genetic code)

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Mentors' Sources:

- AS & A Level Biology Blue Book
- Save My Exams
- Khan Academy